

Mini-Project

CSE350	Data Communication
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Title: Development of a Multi-Campus Subnet-Based University Network Architecture

Description: A Stamford University, like East West University, is a large institution with complex network infrastructure. It provides wired internet access across classrooms, labs, offices, the library, and all campuses. The university also relies on several critical network-based systems, including admissions, advising, results management, eTender, library management, and accounts.

Tasks-

Your task is to design a complete model of a complex network reflecting the university's structure and facilities.

Features:

1. The university will host a central Web Server containing its website.
 - The website must be accessible from every campus using the URL:
like: <http://www.university.edu.bd>
2. A **DNS Server** will be included in any one campus so that people from every campus can browse the university's website.
3. A **DHCP Server** will also be installed in each campus (not a single central DHCP), so that IP addresses are automatically assigned to the hosts of that specific campus.
4. Create at least four LANs (one for each campus).
 - Each LAN must include wired hosts.
5. The university's complete network will cover **Five campuses with five routers**. Connectivity between all hosts across different campuses must be established.
6. Use a **Static routing protocol** for routing.
 - Add the routing tables (with syntax) for all routers in your report.
7. All IP addressing should follow a **Class C scheme** (e.g., 190.150.30.0), with subnetting as needed. Demonstrate **successful ping operations** between at least three LANs to verify connectivity. The web server can host a web page where you may choose to display personal information. However, it must include your full name and student ID.

Project Report will include:

1. Problem Statement
2. Design
3. Experimental Results
4. Conclusion

 **Special note:** Keep your own design with yourself; do NOT distribute your design. Resemblance will be penalized.

Thank you.